Mining Psychological Diagnoses to Identify Vulnerable Populations in Chihuahua, Mexico

Project Proposal

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ABSTRACT

Mental health vulnerability in northern Mexico remains poorly understood despite alarming indicators. Chihuahua, the state with the highest suicide rate in the country, ranks highly on well-being indices while simultaneously exhibiting severe psychological distress. This project analyzes 18,220 psychological consultation records obtained from the Chihuahua government, covering patient demographics and diagnoses. By applying correlation matrices, clustering techniques, and hypothesis testing to identify vulnerable population clusters and diagnostic trends, the analysis aims to inform priority allocation of mental health resources. Preliminary results are expected to reveal demographic and diagnostic patterns that can guide targeted prevention efforts in a region where aggregate prosperity masks hidden psychological suffering.

1 INTRODUCTION

Chihuahua city is the capital of the state with the same name, located in northern Mexico and bordering the US. Although it ir one of the top states in the Better Life Index, it is also the one with the highest suicide rate. The rate in the country is 6.4 per 100,000 persons, while in Chihuahua it more than doubles to 13.7 per 100,000. This signals a hidden vulnerability in the population, where although the general population appears healthy on aggregate indicators, specific subgroups may be disproportionately vulnerable. This project I will analyze about 18,000 psychological diagnoses of Chihuahua's citizens to find trends between disorders.



Figure 1: Mexico, Chihuahua highlighted. Map source: Wikimedia Commons

2 RELATED WORK

There have been 2 main studies about mental health in Chihuahua, specifically for suicide. One study by Selene Oropeza looked at patients admitted at a Chihuahua hospital for self-harm injuries. She analyzed trends such as sobriety, physical illnesses and psychiatric disorders. The last of these should be differentiated from psychological diagnosis, as while disorders describe symptoms or behaviors, on the record of psychological diagnosis a source of origin for the problem is also usually included. Therefore, we can use this previous information to compare whether the same kind of groups suffering from suicidal ideation are the ones seeking professional psychology attention, and where differences lie.

The second study by Montserrat Fernández-López et al. measures the seasonality of suicide, finding that each 1°C rise in temperature was associated with a 2.8% increase in the suicide rate from 2008 to 2018. Since our chosen database includes month of consultation we could also determine if there is an increase in patients seeking help along these lines.

3 PROPOSED WORK

The objective of this project is to mine psychological consultations and diagnoses of Chihuahua's citizens to yield patterns. Some of the results we hope to find are vulnerable population clusters that can be targeted as a priority for the allotment of resources.

The data set used for this project can be acquired from the Chihuahua government website, under the freedom of information act. It contains 18,220 observations with demographic information of patients and their diagnoses. The data has some classification errors and will need preprocessing. Examples of these are inconsistent diagnostic codes, missing demographic fields, and duplicate entries.

The toolset used will mainly be Positron through the R language. The techniques applied through it will be correlation matrices and mean hypothesis testing among others. It is also expected that programming packages such as Stats will be used to fit a K-Means model of clusters and FactoMineR for a PCA projection of them, with Plotly supporting visualization.

4 EVALUATION

The significance threshold that will be used to determine whether we found statistically relevant differences for estimation and inference methods such as population means hypothesis testing, will be the standard alpha = 0.05. Furthermore, since this is an exploratory analysis without preplanned hypothesis, we will conduct our post-hoc analysis of Anova relationships with Bonferroni's adjustments. Clusters modeled will be validated through silhouette scores.

5 TIMELINE AND DISCUSSION

The Project timeline will be as follows:

1. Project proposal - Done
2. Data mining - Done
3. Data preprocessing - Done
4. Analysis – 23rd to 27th of September
5. Visualizations – 28th of September to 4th of October
6. Final report and Presentation – 4th to 11th of October

Some of the potential challenges in this project are in the sensitivity nature of the exploration. Ethical consideration must be taken to not stigmatize any particular protected identity and avoid bias. Additionally, our analysis may be incomplete because it only includes those that sought help and were able to acquire it so it may miss the most vulnerable groups that we are trying to find or make our sample have a different distribution than the actual population.

CONCLUSION

This project proposes a data-driven assessment of psychological diagnoses in Chihuahua, Mexico, a state marked by a striking contradiction: high scores on well-being indices alongside the nation's highest suicide rate. By mining 18,220 clinical consultation records, we aim to uncover demographic and diagnostic patterns that identify vulnerable subgroups and inform resource allocation. The analysis builds on prior research on suicidal ideation and suicide seasonality, extending it by examining who actually seeks psychological care and for what conditions. Methodologically, the project employs clustering, correlation analysis, and hypothesis testing with conservative post-hoc corrections, while acknowledging important limitations: the data reflects only those who sought and obtained care, potentially missing the most at-risk populations, and the sensitive nature of the subject demands careful ethical handling to avoid stigmatization. Despite these constraints, the findings are expected to provide actionable insights for mental health planning in Chihuahua and to demonstrate the value of diagnostic record mining in regions where aggregate indicators conceal serious underlying vulnerability. Future work could integrate external data sources such as emergency room records to capture populations absent from clinical datasets.

REFERENCES

[1] Fernández-López M, Hernández-Montes RI, Álvarez RS, et al. Tendency and seasonality of suicide in Chihuahua, México. A retrospective analysis from 2008 to 2018. Salud Mental. 2021;44(2):43-52.

[2] Selene Oropeza Acevedo. Factores de riesgo asociados al intento suicida en el área de urgencias del HGR no.1 Chihuahua. Thesis, Universidad Autónoma de Chihuahua. 2025

[3] Instituto Municipal de Prevención y Atención a la Salud, Atención Psicológica. https://datos.mpiochih.gob.mx/dataset/atencion-psicologica